

Chapter 8 Aldehydes, Ketones and Carboxylic Acids

NEET UG
CHEMISTRY

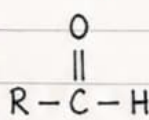
Introduction

These are important carbonyl compounds ($C=O$) containing oxygen.

They show a variety of chemical reactions and are widely found in nature and used in industries.

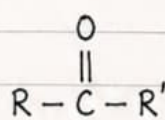
Functional Groups

Aldehydes



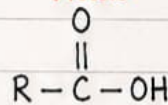
(-CHO)

Ketones



(>C=O)

Carboxylic Acids

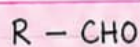


(-COOH)

Classification

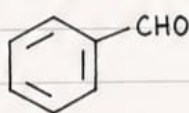
Aldehydes :

(i) Aliphatic Aldehydes :



Ex - CH_3CHO (Ethanal)

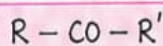
(ii) Aromatic Aldehydes :



Ex - (Benzaldehyde)

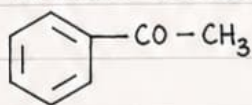
Ketones :

(i) Aliphatic Ketones :



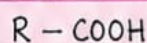
Ex - CH_3COCH_3 (Propanone)

(ii) Aromatic Ketones :



Ex - (Acetophenone)

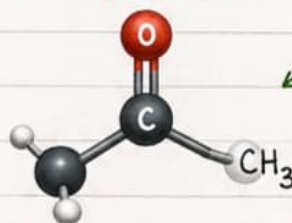
Carboxylic Acids :



Ex - CH_3COOH (Ethanoic acid)

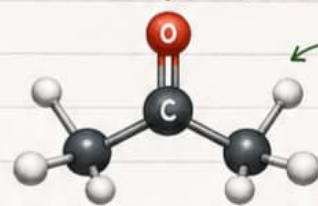
3D Structure & Shape

Aldehyde (e.g., Ethanal)



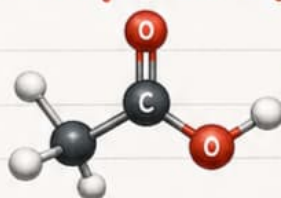
Trigonal planar
 $\angle \approx 120^\circ$

Ketone (e.g., Propanone)



Trigonal planar
 $\angle \approx 120^\circ$

Carboxylic Acid (e.g., Ethanoic acid)



Trigonal planar at carbonyl C
 $\angle \approx 120^\circ$

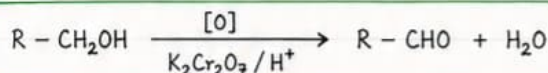
Physical Properties

Property	Aldehydes	Ketones	Carboxylic Acids
1. General formula	$R-CHO$	$R-CO-R'$	$R-COOH$
2. Polarity	Polar	Polar	Highly Polar
3. Boiling point	Lower than acids, higher than hydrocarbons	Slightly higher than aldehydes	Highest (due to H-bonding dimer)
4. Solubility in water	Lower members soluble	Lower members soluble	Lower members highly soluble
5. Odour	Pungent	Pleasant (fruity)	Pungent (acetic acid has strong smell)

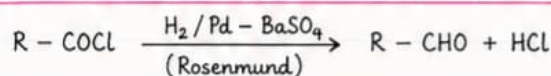
Preparation Methods

A. Aldehydes

1) By oxidation of primary alcohols :

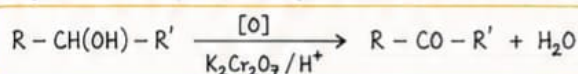


2) By Rosenmund reduction of acid chlorides :

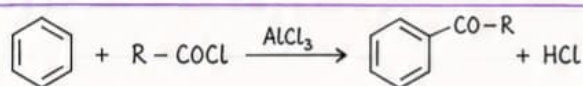


B. Ketones

1) By oxidation of secondary alcohols :

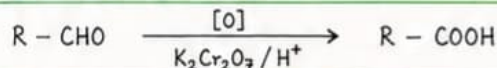


2) By Friedel-Crafts acylation :

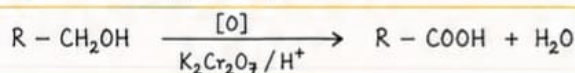


C. Carboxylic Acids

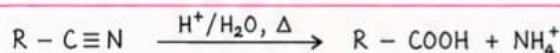
1) By oxidation of aldehydes :



2) By oxidation of primary alcohols :

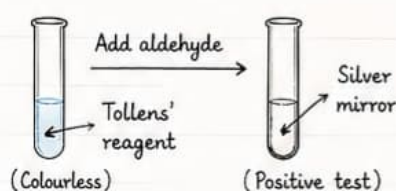
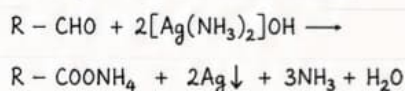


3) By hydrolysis of nitriles :

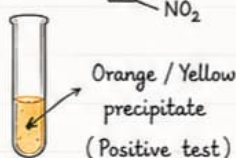
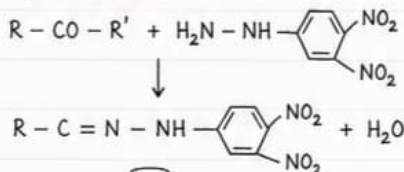


Tests to Identify

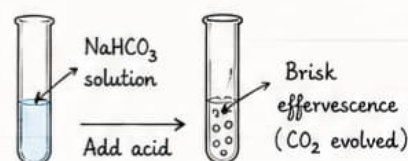
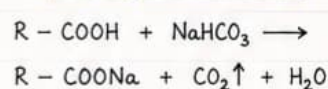
Aldehyde Test - Tollens' test



Ketone Test - 2,4-DNP test



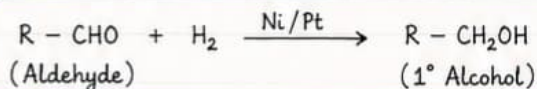
Carboxylic Acid Test



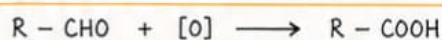
Chemical Properties

A. Aldehydes

1) Addition of Hydrogen (Reduction) :

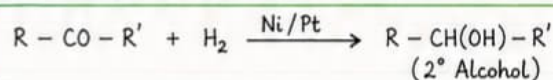
2) Nucleophilic addition with CN^- :

3) Oxidation :



B. Ketones

1) Addition of Hydrogen (Reduction) :



2) Nucleophilic addition (HCN) :

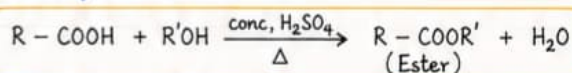
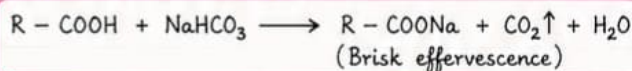


C. Carboxylic Acids

1) Acidic nature :



2) Esterification :

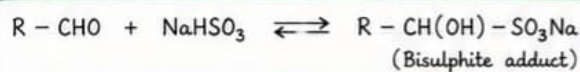
3) Reaction with $NaHCO_3$:★ Key
Points

- Aldehydes are **more reactive** than ketones.
- Carboxylic acids are **more acidic** than alcohols and phenols.

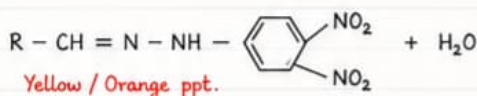
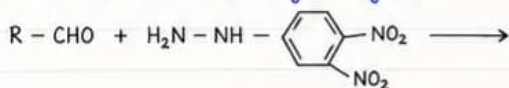
- $-CHO$, $>C=O$ and $-COOH$ are functional groups of this chapter.

Remember
Practice
Succeed

Reactions of Aldehydes

1) With NaHSO_3 (Addition):

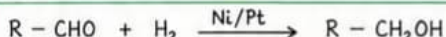
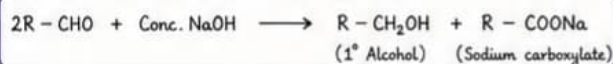
2) With 2,4-DNP (Brady's reagent):



3) Oxidation:

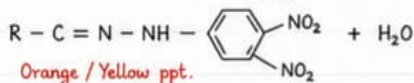
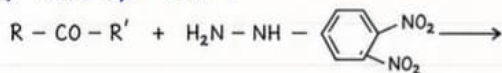


4) Reduction:

(Aldehyde $\rightarrow$ 1° Alcohol)5) Cannizzaro Reaction (No $\alpha\text{-H}$):

Reactions of Ketones

1) With 2,4-DNP:



2) Reduction:

(Ketone $\rightarrow$ 2° Alcohol)

3) No Cannizzaro Reaction

(Ketones do not have $-\text{CHO}$ group)

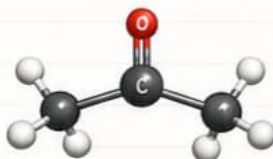
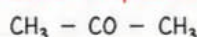
3D Structures (Ball & Stick Models)

Aldehyde (Ethanal)



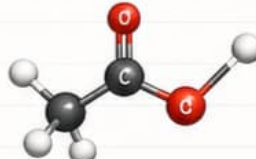
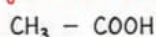
Trigonal planar at $-\text{CHO}$
($\text{C}=\text{O}$ bond length $\approx 1.21 \text{ \AA}$)

Ketone (Propanone)



Trigonal planar at $\text{C}=\text{O}$
($\text{C}=\text{O}$ bond length $\approx 1.21 \text{ \AA}$)

Carboxylic Acid (Ethanoic acid)



Trigonal planar at carbonyl C
($\text{C}=\text{O}$ bond length $\approx 1.21 \text{ \AA}$)

Mnemonics



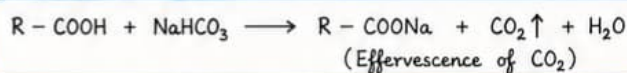
Aldehyde: "Aldehydes Add -H easily."
(Easily oxidised)

Ketone: "Ketones Keep Calm!"
(Less reactive)

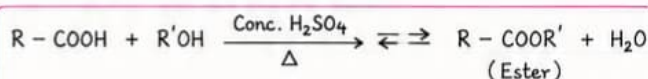
Acid: "Acids React with Bases
and Make Salt."



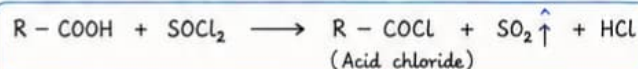
Reactions of Carboxylic Acids

1) With NaHCO_3 :2) With NaOH (Neutralisation):

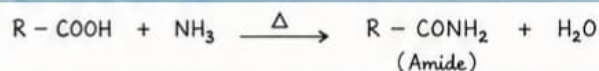
3) Esterification (Fischer method):



4) Acid chloride formation:

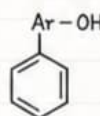
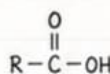


5) Amide formation:



Acidity Order

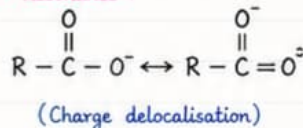
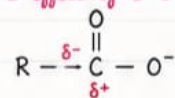
Carboxylic acids > Phenols > Alcohols > Water



Why Carboxylic Acids are more acidic?

Conjugate base of carboxylic acid (RCOO^-) is stabilised by

• Resonance:

• -I effect of $\text{C}=\text{O}$:

stabilises the anion.

★ Quick Revision

- Aldehydes have $-\text{CHO}$ group, give **positive Tollens' test**.
- Ketones have $>\text{C}=\text{O}$, give 2,4-DNP test (orange ppt.).
- Carboxylic acids are more acidic than alcohols & phenols.
- $-\text{COOH}$ group shows neutralisation, esterification & formation of derivatives.

Uses in Daily Life & Industry

★ Aldehydes:

- Formaldehyde (methanal) $\rightarrow$ Disinfectant, preservative, in making resins (Bakelite).
- Benzaldehyde $\rightarrow$ Flavouring agent in food and perfumes.



★ Ketones:

- Acetone (propanone) $\rightarrow$ Solvent in nail polish remover, paints, varnishes.
- Used in production of chloroform, bisphenol-A, medicines.

★ Carboxylic Acids:

- Acetic acid $\rightarrow$ Vinegar (4-8%), food preservative.
- Benzoic acid $\rightarrow$ Preservative in pickles, jams.
- Used in soaps, detergents, polymers, drugs, fragrances.

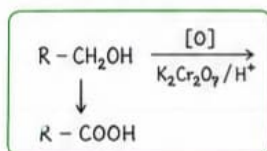


Comparison of Aldehydes & Ketones

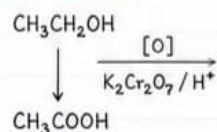
Basis	Aldehydes	Ketones
Functional group	R - CHO (Terminal)	R - CO - R' (Internal)
Position of C = O	At the end of the carbon chain.	In the middle of the carbon chain.
Oxidation	Easily oxidised to carboxylic acids.	Do not oxidise easily.
Tollens' test	Positive (Silver mirror formed)	Negative
2,4-DNP test	Positive	Positive
Examples	HCHO, CH ₃ CHO, C ₆ H ₅ CHO	CH ₃ COCH ₃ , C ₆ H ₅ COCH ₃

Preparation of Carboxylic Acids (Important Methods)

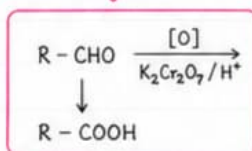
1) Oxidation of 1° Alcohols



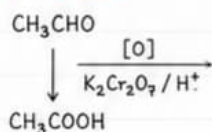
Ex - Ethanol



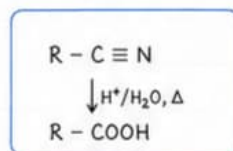
2) Oxidation of Aldehydes



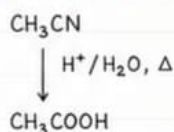
Ex - Ethanal



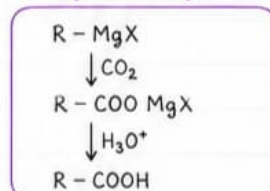
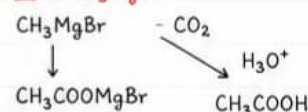
3) Hydrolysis of Nitriles



Ex - Acetonitrile



4) Carboxylation of Grignard Reagent

Ex - CH₃MgBr

Derivatives of Carboxylic Acids

Derivative	General Formula	Prepared from	Example	Use
Acid Chlorides	R - COCl	R - COOH + SOCl ₂	CH ₃ COCl (Acetyl chloride)	Organic synthesis, to prepare esters, amides.
Acid Anhydrides	(R - CO) ₂ O	2RCOOH $\xrightarrow[\Delta]{\text{P}_2\text{O}_5}$	(CH ₃ CO) ₂ O (Acetic anhydride)	Acetylation, organic synthesis.
Esters	R - COOR'	R - COOH + R'OH $\xrightarrow[\text{conc. H}_2\text{SO}_4]{}$	CH ₃ COOC ₂ H ₅ (Ethyl acetate)	Solvents, flavours, fragrances.
Amides	R - CONH ₂	R - COOH + NH ₃ $\xrightarrow[\Delta]{}$	CH ₃ CONH ₂ (Acetamide)	In polymers, medicines.
Nitriles	R - C $\equiv$ N	R - COOH $\xrightarrow{\text{P}_2\text{O}_5}$	CH ₃ CN (Acetonitrile)	Solvents, in synthesis.

Summary

- ✓ Aldehydes have -CHO group and are easily oxidised.
- ✓ Ketones have >C=O group and are less reactive.
- ✓ Carboxylic acids have -COOH group and are acidic.
- ✓ Aldehydes give Tollens' positive test.
- ✓ Carboxylic acids form many derivatives useful in industries & daily life.

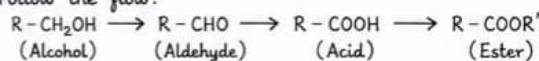


One-Liners

- Aldehydes are more reactive than ketones.
- Carboxylic acids are more acidic than phenols.
- CHO gives Tollens' test; >C=O does not.
- Acids reduce NaHCO₃ with brisk effervescence.
- Ester smells $\rightarrow$ sweet & fruity.

Exam Tip

Follow the flow:



Remember reagents & test to score high in NEET!

